

Labsheet 1

Topic: Distributed Applications using Sockets(Chat Program)

Task 1: Server

```
C:\Users\garla\Desktop\Lab Sheets\Distributed Systems>python server_task1.py
Chat Server started...
Waiting for client...
Connected to client: ('127.0.0.1', 54497)
Client: hello
Server: Hi client
Client disconnected.
Server closed.
```

Client:

```
C:\Users\garla\Desktop\Lab Sheets\Distributed Systems>python client_task1.py
Connected to server.
Client: hello
Server: Hi client
Client: exit
Connection closed.
```

Task 2:

Server

```
C:\Users\garla\Desktop\Lab Sheets\Distributed Systems>python server_task2.py
[SERVER STARTED] Waiting for connections...
[NEW CONNECTION] ('127.0.0.1', 52554) connected.
[ACTIVE CONNECTIONS] 1
Client ('127.0.0.1', 52554): Hello server..
Reply to ('127.0.0.1', 52554): hii, client1
[DISCONNECTED] ('127.0.0.1', 52554)
[NEW CONNECTION] ('127.0.0.1', 52555) connected.
[ACTIVE CONNECTIONS] 1
Client ('127.0.0.1', 52555): Hi server..
Reply to ('127.0.0.1', 52555): [NEW CONNECTION] ('127.0.0.1', 51152) connected.
[ACTIVE CONNECTIONS] 2
Hii client2
Client ('127.0.0.1', 51152): Hii server
Reply to ('127.0.0.1', 51152): hii client3
[DISCONNECTED] ('127.0.0.1', 51152)
[DISCONNECTED] ('127.0.0.1', 52555)
[NEW CONNECTION] ('127.0.0.1', 55959) connected.
```

Client1:

```
C:\Users\garla\desktop\Lab Sheets\Distributed Systems>python client_task2.py
Connected to server.
Client: Hello server..
Server: hii, client1
Client: exit
Connection closed.
```

Client2:

```
C:\Users\garla\desktop\Lab Sheets\Distributed Systems>python client_task2.py
Connected to server.
Client: Hi server..
Server: Hii client2
Client: exit
Connection closed.
```

Client 3:

```
C:\Users\garla\desktop\Lab Sheets\Distributed Systems>python client_task2.py
Connected to server.
Client: Hii server
Server: hii client3
Client: exit
Connection closed.
```

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Connecte